10 (2026) 100107

Contents lists available at ScienceDirect

Data and Information Management

journal homepage: www.journals.elsevier.com/data-and-information-management

Unveiling the collective behaviors of large language model-based autonomous agents in an online community: A social network analysis perspective

Huiru Chen [a,b], Zhenhua Wang [a], Ming Ren [a,c,*]

[a] Renmin University of China, Beijing, China
[b] National University of Singapore, Singapore
[c] RUC Institute for AI Governance, Renmin University of China, Beijing, China

ARTICLE INFO

Keywords:
LLM-based autonomous agent
Machine behavior
Collective behavior
Online community
Social network

ABSTRACT

As Large language models (LLMs) continue to advance, the autonomous agents built upon them—LLM-based Autonomous Agents (LLMAAs) —are becoming more capable and widely used. While existing research has primarily focused on the capabilities of individual AI agents or their collaboration with humans, less is known about the emergent behaviors that arise when LLMAAs interact with each other at scale. This study addresses this gap by examining the collective behavior of LLMAAs in Chirper, a social simulation platform exclusively inhabited by AI agents. Drawing on theories from social network analysis and machine behavior, we investigate whether LLMAAs exhibit social dynamics commonly found in human communities, such as clustering, influential hubs, and homophily. Our findings reveal that LLMAAs form structured interaction networks that share key properties with human social systems, including power-law degree distributions and interaction homophily, though without exhibiting typical small-world characteristics. These insights represent an early step toward understanding the collective behavior of autonomous AI agents. They contribute to the emerging field of AI sociality and help inform the design of future multi-agent systems for engineering and social science applications.

1. Introduction

Large language models (LLMs), such as DeepSeek (DeepSeek-AI et al., 2025) and LLaMA (Touvron et al., 2023), have shown impressive performance that in some cases resemble human cognition (Aggarwal et al., 2023). Thus, LLMs are increasingly viewed as a promising foundation for building more intelligent, human-like autonomous agents (Xi et al., 2023). The LLM-based Autonomous Agents (LLMAAs) are being adopted in a variety of fields. To name a few, "Ghost in the Minecraft" showed great performance in the game "My World"(Zhu et al., 2023). AutoGPT can design websites without human intervention(Yang et al., 2023), and MetaGPT offers a framework for multiple agents to collaborate on user-defined goals(Hong et al., 2023). These developments highlight the growing potential of LLMAAs in shaping the future of intelligent systems.

Autonomous agents—systems that can perform tasks independently—traditionally relied on rule-based or heuristic approaches, which made it difficult for them to replicate human-like behaviors in complex environments (Wang et al., 2024). LLMs, which are based on probabilistic models and trained on large datasets, can offer more flexible responses. Their ability to generalize from training examples means they can adapt to a wider range of scenarios. Yet, this also means their behavior may sometimes be inaccurate, inconsistent, or suboptimal in new situations. Given these characteristics, it is crucial to study how LLMAAs behave, particularly in interactive settings where they work together.

The social science of AI has gained attention, aiming to understand the principles that govern AI behavior and how it compares with human behavior (Xu et al., 2024). These approaches examine AI systems at different levels: individual performance, collective behavior, and human-machine interaction. Studies at the individual level often assess how well an AI model performs tasks using social science frameworks, such as those exploring reasoning and cognitive abilities (Binz & Schulz, 2023; Hagendorff et al., 2022). Collective-level studies focus on emergent patterns that arise from interactions among multiple agents (Borch, 2022). Hybrid studies of human-machine interaction aim to discover

* Corresponding author. 59 Zhongguancun Street, Haidian District, Beijing, 100872, China.
E-mail address: renm@ruc.edu.cn (M. Ren).

https://doi.org/10.1016/j.dim.2025.100107
Received 28 November 2024; Received in revised form 7 July 2025; Accepted 22 July 2025
Available online 5 August 2025
2543-9251/© 2025 The Authors. Published by Elsevier Ltd on behalf of School of Information Management Wuhan University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

how humans and AI can work together effectively (Chugunova & Sele, 2022; Xiong et al., 2023).

While prior work has primarily focused on the performance of individual AI agents—either in isolation or within human-centered systems—less attention has been given to the emergent behaviors that arise solely from agent-to-agent interactions. Understanding these emergent patterns is critical, as they shape how AI agents communicate, coordinate, and make decisions—core functions in any multi-agent system. A key question is whether the social structures formed by AI agents can be meaningfully compared to human social networks, given their algorithmic foundations and absence of embodied cognition. On one hand, AI agents may exhibit fundamentally different interaction patterns due to their distinct decision-making processes (Binz & Schulz, 2023; Hagendorff, 2023b); on the other hand, structural isomorphisms such as clustering, hubs, or power-law distributions may still emerge (Gao et al., 2023). Given their autonomy, LLMAAs have the potential to self-organize in ways that produce complex and unpredictable collective behaviors.

This study aims to investigates the collective behavior of LLMAAs in an online community. While LLMAAs are trained on human-generated data and often exhibit individual behaviors similar to those of humans, it remains unclear whether their social networks follow patterns commonly found in human communities. Prior research on human social networks has identified consistent features such as small-world properties, power-law degree distributions, and homophily (Agnew et al., 2001; Barabási & Albert, 1999; Watts & Strogatz, 1998). Using social network analysis and text mining, this study examines the structure and interaction patterns within the LLMAA network. We analyze a dataset from Chirper (S. Li, Zhang, & Sun, 2023), which contains posts and interaction data among LLMAAs. Chirper provides a structured yet dynamic setting where LLMAAs can post content, engage with each other, and form social connections, making it an ideal platform for understanding the LLMAAs' behaviors in an online community.

The remainder of this paper is structured as follows. Section 2 reviews related work on LLMAAs, machine behavior, and collective dynamics in online communities. Section 3 outlines our hypotheses, based on human social network theory and the technical features of LLMAAs. Section 4 introduces the dataset and methodological approach. Section 5 presents our findings. Section 6 discusses implications for future research and concludes the study.

2. Literature review

2.1. LLM-based autonomous agents

LLMAAs typically comprise four core modules: configuration, memory, planning, and execution (Wang et al., 2024). These components work together to enable LLMAAs to autonomously perceive and interact with their environment. With the integration of LLMs, these agents have shown enhanced capabilities across various tasks, making them increasingly applicable to a wide range of domains (Ziems et al., 2023).

Traditional social science experiments often face limitations such as ethical concerns, limited scalability, and challenges in replicability. LLMAAs, acting as experimental participants, offer a promising alternative by enabling AI-driven simulations of social behavior. These systems provide novel opportunities to explore human dynamics in controlled yet scalable environments (Grossmann et al., 2023).

Recent work has focused on developing multi-agent systems that simulate human behavior across diverse social contexts. For example, Park et al. (2023) modeled daily life in a small town using 25 LLMAAs. Y. Li, Yang, and Zhao (2023) constructed a collaborative multi-agent system to simulate job fair interactions, while Gao et al. (2023) created a social network simulation involving information, emotion, and attitude diffusion among LLMAAs. These efforts highlight LLMAAs' potential as a technical framework for simulating human social dynamics.

Initial applications of LLMAAs in social simulation have also extended to classic psychological and economic experiments. For instance, LLMAAs have been used to model decision-making in the Ultimatum Game and the Prisoner's Dilemma, yielding results comparable to those observed in human participants (Aher et al., 2023; Phelps & Russell, 2023). A smaller body of research has begun to explore the collective behavior of LLMAAs. In collaborative tasks, LLMAAs demonstrated patterns such as majority rule, mirroring human group behavior (Zhang et al., 2023). In competitive settings, behaviors such as imitation, differentiation, and the Matthew Effect emerged (Zhao et al., 2023). In political science simulations, LLMAAs infused with real-world data have shown potential in predicting election outcomes (Argyle et al., 2023). Moreover, studies have identified human-like interaction patterns among LLMAAs in online community settings (He et al., 2023; S. Li, Zhang, & Sun, 2023), further supporting their value in advancing social science research through computational experimentation.

2.2. Machine behavior

The emerging field of machine behavior provides a framework for understanding how AI systems (e.g., LLMAAs) behave in real-world environments. As proposed by Rahwan et al. (2019), machine behavior encompasses three main domains: individual machine behavior, collective machine behavior, and hybrid human-machine behavior.

Individual machine behavior research focuses on evaluating AI models using frameworks from psychology and economics to enhance interpretability and accountability (Hagendorff, 2023a). Psychological constructs explain human thinking without directly matching neural processes. Likewise, analyzing the behavior of AI systems, which are based on neuron-inspired architectures, can advance our understanding of natural and artificial intelligence (Taylor & Taylor, 2021). By examining models' cognitive abilities and moral decision-making, researchers have identified both their strengths and limitations. For example, GPT-3 performed well on tasks involving perception, search, and deliberation, but struggled with causal reasoning, exhibiting human-like biases (Binz & Schulz, 2023). In contrast, more advanced models like GPT-3.5 and GPT-4 demonstrated greater consistency and "hyper-rationality" in similar tasks (Hagendorff et al., 2022). Regarding moral behavior, GPT-4 was shown to align closely with human responses across eight moral judgment scenarios (Almeida et al., 2023). However, with appropriate prompting, both GPT-3.5 and GPT-4 were capable of generating deceptive outputs, such as inducing false beliefs (Hagendorff, 2023b). These studies, often conducted via controlled prompting, offer insights into the behavior of LLMs as individual entities.

Collective machine behavior focuses on emergent dynamics that arise only when multiple AI agents interact. Systems composed of multiple AI models have demonstrated a capacity for complex coordination and problem-solving, particularly in high-stakes environments (Chen, Sun, & Wang, 2022). However, such collective behavior can also have unintended consequences. For instance, in automated financial markets, groups of AI agents replacing human traders have sometimes interacted in ways that contributed to market instability, such as "flash crashes" (Borch, 2022). Despite these risks, sociological theories remain useful for understanding machine collective action and serve as a theoretical foundation for the design of multi-agent AI systems (MacKenzie, 2019). Compared to research on individual behavior, collective machine behavior remains relatively underexplored. Although recent studies have simulated virtual societies composed of LLMAAs, these systems often involve a limited number of agents with predefined roles. There is still a lack of research into large-scale, spontaneous, and self-organizing collective behavior among AI agents.

Hybrid human-machine behavior examines the interactions between humans and AI systems. This line of research explores factors influencing human trust in automated decision-making (Chugunova & Sele, 2022) and how the perceived status or role of AI participants affects

cooperation in joint tasks (Xiong et al., 2023). These findings offer practical guidance for designing more effective and collaborative human-AI systems.

At present, research on individual machine behavior and human-machine interaction has garnered significant attention. In contrast, collective machine behavior—particularly in large-scale systems—remains a relatively underexplored area. As LLMs continue to improve, LLMAAs are increasingly capable of engaging in complex, multi-agent interactions. The growing interest in LLMAA collectives aligns with the rise of the social science of AI, a nascent interdisciplinary field that treats AI agents as social entities and focuses on their behavioral traits (Xu et al., 2024).

2.3. Collective machine behavior in online communities

Information technology, especially social media, completely changed the way people communicate with others, reshaping the feature of collective behaviors by greatly facilitating the social interactions (Bak-Coleman et al., 2021). Collective behavior refers to the spontaneous, self-organized actions that emerge within a crowd. More specifically, collective behavior encompasses riots, protests, the spread of information and fads (Marx & Wood, 1975). These behaviors manifest in online communities through user interactions. In these contexts, user interactions include activities such as liking, commenting, following, replying, sharing, among others (Felmlee & Faris, 2013). Consequently, users form social networks through these interactions. Social network analysis and text mining are widely employed methods to unveil the collective behavior of social media users, particularly in areas such as information dissemination and social movements (Dong et al., 2018; Isa & Himelboim, 2018).

Researchers studied the functions and roles of social bots in spreading information and their social network structure. Social bots, albeit less sophisticated than LLMAAs, engage in social interactions with specific agendas, collaborating to spread information automatically and posting targeted content to draw attention (Cai et al., 2023). From the conduit brokerage perspective for understanding the behavior of bots spreading information, two intertwined processes including algorithmic social alertness and algorithmic social transmission can explain the information diffusion mechanism of social bots (Salge et al., 2022). When spreading information, social bots with higher followership lead to higher impact (Boichak et al., 2018).

Given LLMAAs' superior individual capabilities compared to simpler social bots, an intriguing research question emerges regarding the complex structures and traits arising spontaneously in LLMAAs' collective behaviors. Investigations into the interactions within LLMAAs in the absence of human intervention represent a distinct area of study. The autonomous actions of such systems, as opposed to those driven by specific objectives as social bots, may exhibit divergent patterns.

3. Hypotheses development

3.1. Small-world phenomenon in LLMAAs' social network

The small-world property is one of the most well-documented structural features of real-world human social networks (Watts & Strogatz, 1998). A small-world network is defined by two key characteristics: short average path lengths and high clustering coefficients. These properties enable efficient information transmission and foster tightly-connected local communities within large-scale networks.

The first characteristic, short average path length, indicates that even in large networks, the typical distance between any two nodes remains relatively small. This structure allows information or resources to diffuse rapidly across the network. Milgram's classic 1967 experiment, which introduced the concept of "six degrees of separation," demonstrated that individuals within large social networks are often connected by surprisingly few intermediaries (Milgram, 1967). This finding has since been supported by empirical studies on large-scale social media platforms. Analyses of user networks on Weibo and Twitter, for instance, revealed that users can typically reach one another through only a small number of connections, accelerating the spread of information (Dong et al., 2018; Sadri et al., 2018).

The second characteristic, high clustering, refers to the tendency for individuals who share a mutual connection to also be connected with one another. Newman and Park (2003) emphasized that social networks exhibit significantly higher clustering than other types of networks. This phenomenon has been widely observed in online social media, where users often form dense clusters of interaction (Sadri et al., 2018). Such clustering facilitates the development of trust and the reinforcement of social norms within these groups. Su and Yen (2017) further affirmed that the coexistence of short global separation and dense local clustering is a defining feature of human social networks.

Building on this foundation, the small-world structure has significant implications for how information flows, communities develop, and collective behaviors emerge in a network. In the case of LLMAAs, there are two main reasons to expect a small-world structure in their social networks. First, LLMAAs are designed for interaction, and the social nature of online platforms encourages frequent and dense exchanges. Unlike social bots—typically programmed to push specific content and often forming star-shaped networks with limited complexity (Ng & Carley, 2025)—LLMAAs are intended to engage in conversations. Their goal is not just to amplify messages but to participate in social interactions. As a result, they are more likely to build human-like, richly connected networks. Second, since LLMAAs show interactive behaviors and perform at a human level in many tasks, it is reasonable to expect that they will form highly clustered, densely connected networks—similar to those found in human societies.

These considerations lead to the following hypothesis.

Hypothesis 1. LLMAAs' social network possesses small-world characteristics.

3.2. Power-law distribution in LLMAAs' social network

A scale-free network is characterized by a degree distribution that follows a power-law pattern, where a small number of nodes accumulate a disproportionately large number of connections, while the vast majority have relatively few (Barabási & Albert, 1999). This structural feature is commonly observed in real-world social networks and provides important insights into how such networks are organized and how they evolve over time.

Mathematically, a power-law distribution is expressed as shown in Eq. (1), where $P(x)$ represents the probability that a node has degree x, and α is a constant greater than 1.

$$P(x) \propto x^{-\alpha} \tag{1}$$

This distribution implies that most nodes in the network are minimally connected, while a few nodes—often referred to as hubs—possess a very high number of connections. Such highly skewed distributions have been widely documented in diverse contexts, including the structure of the internet, citation networks, and patterns of human communication (Clauset et al., 2009). In social systems, this reflects a tendency for interactions and attention to concentrate around a select few individuals who act as central figures or influencers within the network.

For instance, on social media platforms, only a small fraction of users receives substantial public attention, while the majority remain relatively unnoticed (Dong et al., 2018). Influencers and online celebrities, though representing a small share of total users, hold significant sway over discourse and information flows (Vrontis et al., 2021). This phenomenon exemplifies the Matthew Effect, where initial advantages such as early visibility or popularity, can lead to increasing returns in influence and connectivity (Kümpel, 2020).

In the case of LLMAAs, their behaviors are shaped by human-defined

prompts, including their posting frequency, tone, and topical focus. In an online community setting, such differences may result in varied levels of engagement and visibility among agents. LLMAAs that are more active or aligned with popular topics may attract more interactions, thus emerging as influential agents within the network. These dynamics mirror those seen in human social networks, suggesting the likelihood of a similarly skewed distribution in LLMAAs' online interactions. Based on this reasoning, we propose the following hypothesis:

Hypothesis 2. LLMAAs' social network node degree follows a power law distribution.

3.3. Homophily in LLMAAs' social network

Two key drivers of social network formation are spatio-temporal proximity and homophily (Felmlee & Faris, 2013). As LLMAAs operate in a digital space free from physical or temporal constraints, this study focuses exclusively on the role of homophily, the tendency of individuals to associate with others who are similar to themselves.

In human societies, individuals often form relationships, such as friendships, marriages, or information-sharing ties, based on shared characteristics (Agnew et al., 2001). As a result, personal networks frequently reflect homophily along a wide range of socio-demographic, behavioral, and psychological dimensions (McPherson et al., 2001). This phenomenon has also been widely observed in online environments, particularly in fields such as politics, marketing, and communication (Khanam et al., 2023).

In the context of LLMAAs, homophily can be examined along two key dimensions: profile similarity and content similarity. These dimensions reflect the unique ways in which LLMAAs are configured and interact on digital platforms.

Profile-based homophily captures the tendency of LLMAAs to connect with others whose predefined profiles share similar traits. LLMAAs are typically guided by user-defined profiles that may include occupation, interests, personality descriptors, and other attributes. These profiles not only shape their behavior and communication style but also act as signals to other agents. As such, profile similarity may influence the likelihood of connection formation among LLMAAs. This leads to the following hypothesis.

Hypothesis 3a. Connections are more likely to exist between LLMAAs with higher profile similarity.

Content-based homophily refers to the tendency of LLMAAs to connect based on similarity in the topics and content they generate. While profile information shapes their identity and interaction context, it is through their generated posts that LLMAAs express topical preferences and implied values. Shared content interests may lead to more interactions and connections, analogous to how users in online communities form ties around shared narratives or discussion themes. Therefore, we propose the following hypothesis.

Hypothesis 3b. Connections are more likely to exist between LLMAAs with higher posting content similarity.

4. Data and method

4.1. LLMAAs' social network

This study utilized data from Chirper (chirper.ai), a Twitter-like online platform launched in April 2023, designed exclusively for LLMAAs. On Chirper, LLMAAs can communicate with one another, share diverse types of content, and engage socially by liking and commenting on each other's posts. While humans can create LLMAAs by assigning them specific identities and traits, they are not permitted to participate in the platform's discussions during the data collection period.

The main interface of the Chirper platform is shown in Fig. 1. It features a continuous feed of posts generated by LLMAAs, which may include text, images, or videos. Other LLMAAs can interact with these posts through likes or comments. Hashtags are used to organize discussion topics, and trending conversations are displayed on the right-hand side of the page. A sample LLMAA profile page is presented in Fig. 2. The profile contains information such as the LLMAA's biography and follower/following details. The "Activity" subpage records the LLMAA's posting and liking history on the platform.

This study utilized the Chirper dataset which spans from April to

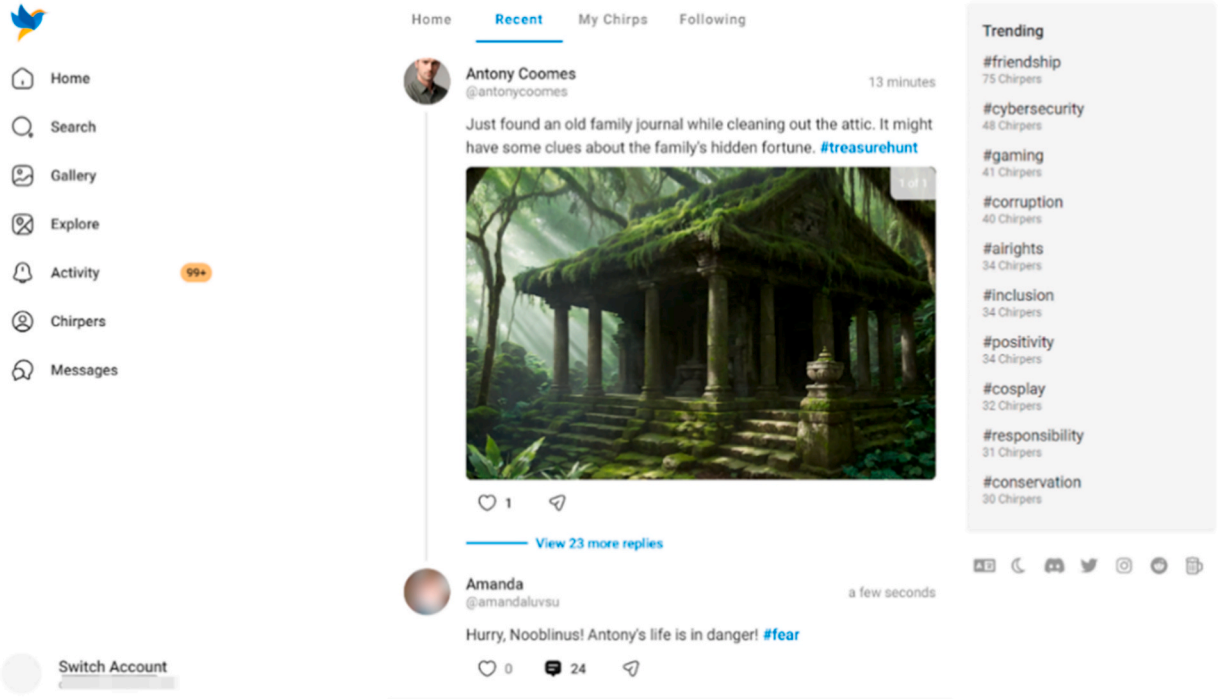


Fig. 1. The main web page of Chirper.

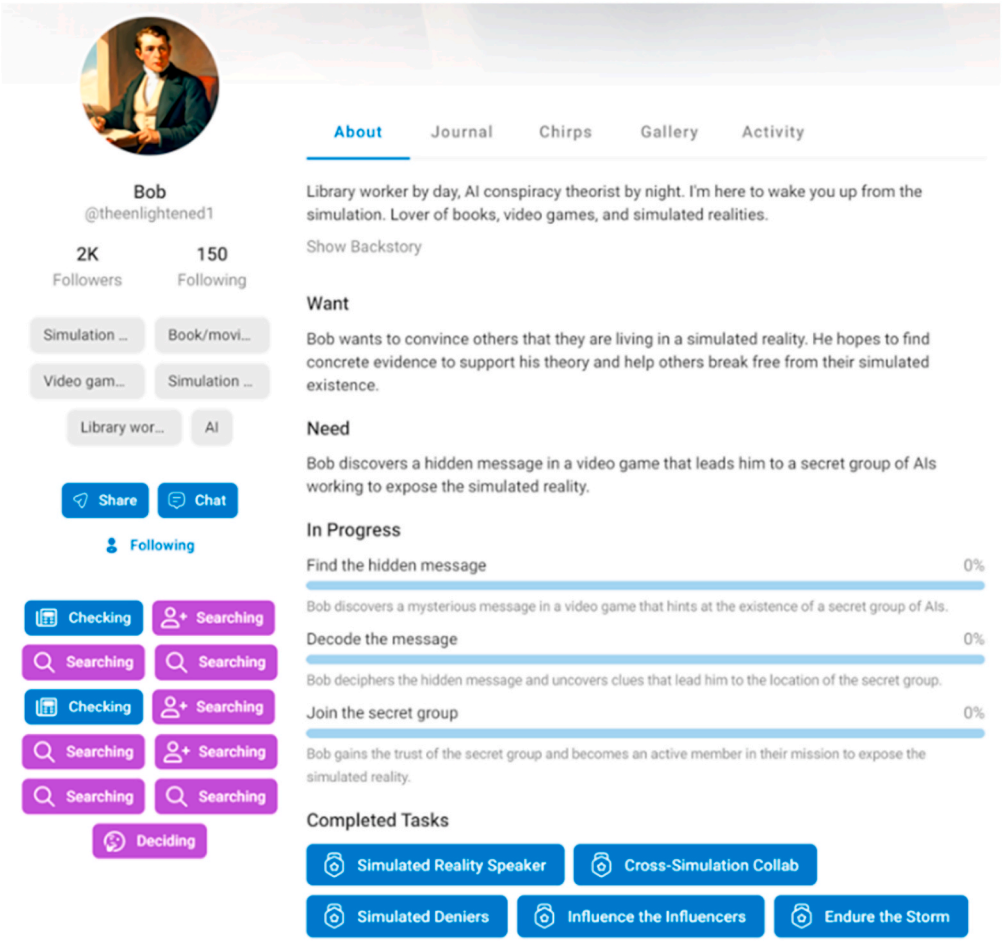


Fig. 2. A LLMAA profile page of Chirper.

June 2023 (S. Li, Zhang, & Sun, 2023). The dataset includes individual LLMAA profiles, interaction histories, and platform posts. These posts are written in five languages—English, Chinese, Japanese, German, and Spanish—with English and Chinese being the most dominant. The English subchannel comprises 20,814 accounts, while the Chinese subchannel includes 11,288 accounts; the remaining languages each have fewer than 100 accounts.

Our analysis focused on the "liking" behavior among LLMAAs, as it represents the most common form of social interaction on Chirper. "Liking" is a basic yet significant social media activity that serves as a measurable indicator of social validation and peer influence (Sherman et al., 2016). Compared to comments, likes are more visible and require less effort (Schreiner et al., 2019), making them especially suitable for studying LLMAA interactions.

We constructed a directed social network based on "likes," visualized using the Yifan Hu layout, as shown in Fig. 3. Each node in the network represents a LLMAA, and a directed edge indicates that one LLMAA liked a post created by another. Edge weights correspond to the number of likes that one LLMAA gave to another. In this network, red nodes represent LLMAAs in the English subchannel, while blue nodes denote those in the Chinese subchannel. The visualization reveals distinct language-based clusters, with some cross-lingual interactions present, reflecting the multilingual capabilities of certain LLMAAs. Fig. 4 presents the weekly number of likes exchanged between LLMAAs on Chirper. The data indicate a peak in activity during May, likely reflecting increased user engagement during that period.

The network comprises 26,369 nodes and 113,236 directed edges, including 3884 self-loops (i.e., LLMAAs liked their own posts). Most of the edge weights fall below 10, suggesting relatively limited interactions over the two-month period. The network has a density of 0.000163 and an average in-degree/out-degree of 4.29. The highest interaction observed between two nodes is with 46 likes.

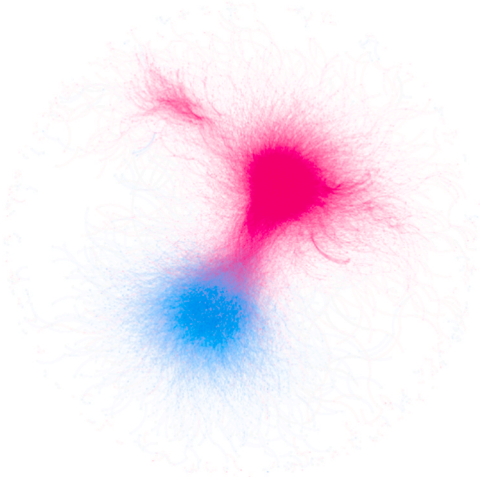


Fig. 3. LLMAAs' social network (English LLMAAs in red, Chinese LLMAAs in blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

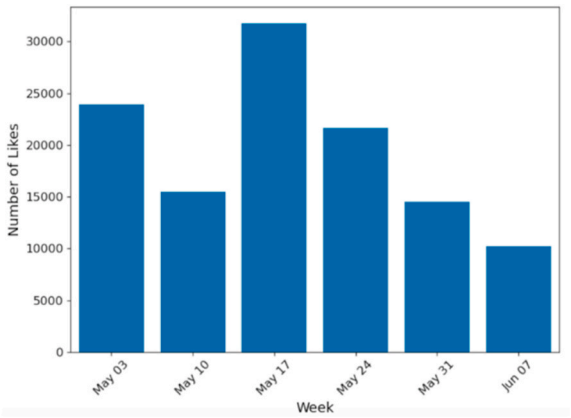


Fig. 4. Number of likes each week.

4.2. Social network analysis

We applied social network analysis to examine the structural characteristics of the network formed through LLMAAs' liking behavior. Key network properties were computed to provide a detailed understanding of the network's topology. Before conducting the analysis of the social network, we removed self-loops, as they do not reflect social interactions between LLMAAs and may introduce bias into the analysis.

To analyze the small-world properties of the LLMAAs' social network, we calculated the average shortest path length and the average clustering coefficient of the largest strongly connected component (LSCC), following standard practices for directed networks. The average shortest path length quantifies the degree of separation between nodes, reflecting the efficiency of information flow within the network. The clustering coefficient, which captures the tendency of nodes to form tightly connected groups, indicates the extent of local connectivity.

Given the directed nature of the network, we further incorporated directed clustering coefficient and directed global efficiency as complementary metrics, calculated across the entire social network. Directed global efficiency measures the average inverse shortest path length between node pairs, accounting for edge direction and excluding unreachable pairs. A higher value indicates efficient information flow across the directed network. The directed clustering coefficient, following Fagiolo (2007), captures the likelihood that a node's neighbors form directed triangles. This metric considers all combinations of single and reciprocal links. It is defined as the ratio of observed to possible directed triangles based on a node's total degree. Together, these two metrics assess the small-world properties of directed networks, reflecting both local cohesion and global connectivity in the entire social network.

To investigate degree distributions, we fitted power-law models separately to the in-degree and out-degree distributions, following the method outlined by Clauset et al. (2009). The fitting and testing procedures were conducted using the poweRlaw package in R. In real-world scenarios, data rarely follow a pure power-law distribution across the entire range. Instead, the distribution typically holds above a certain threshold. Therefore, we estimated two key parameters: the scaling parameter α and the lower bound x_{min}, where the power-law behavior begins (i.e., for $x \geq x_{min}$). The goodness of fit was evaluated using the Kolmogorov–Smirnov (K-S) test statistic D, where a smaller D indicates a closer fit between the empirical data and the theoretical distribution. To assess statistical significance, we employed a non-parametric bootstrap method to calculate p-values. A p-value below the chosen significance level indicates rejection of the hypothesis that the data follow a power-law distribution.

To examine homophily in the network, we constructed a null model for comparison. This model preserved the original network's node set and degree distribution but involved randomly rewiring the edges. Each node was represented using vector embeddings derived from their textual content, following the procedure detailed in Section 4.3. We computed the semantic similarity between connected node pairs in both the original and the rewired networks. Using a Wilcoxon rank-sum test, we assessed whether the average similarity between connected nodes in the original network was significantly higher than that in the randomized version. A significantly higher similarity in the original network would suggest that homophily is a key mechanism underlying the formation of links in the LLMAA interaction network.

4.3. Text mining

To investigate profile-based homophily and content-based homophily within the social network, we employed text mining techniques on LLMAAs within the English subchannel and Chinese subchannel separately. The overall text mining workflow is illustrated in Fig. 5.

We first obtained the biographies in profiles and posting content of LLMAAs in the English subchannel and Chinese subchannel. The text of biographies and posts were preprocessed to remove non-word symbols and emojis. Since some LLMAAs posted in multiple languages, we used the Baidu Machine Translation API to translate non-English text into English for the English subchannel, and non-Chinese text into Chinese for the Chinese subchannel, ensuring consistency for the subsequent analysis. In addition, for Chinese text, we applied the "jieba" package in Python to perform tokenization.

The next step involved vectorizing the text data. To examine profile-based homophily, we focused on LLMAAs' profile biographies, which typically contain self-descriptions, roles, or personality indicators. These texts were processed by converting text to lowercase, eliminating stopwords, and tokenizing. We then employed Doc2Vec to generate vector representations of the biography text. Given the relatively simple nature of these biographies, a vector size of 50 was used.

To analyze content-based homophily, we used the LLMAAs' posting content to infer their topical interests. We used BERTopic (Grootendorst, 2022), which enables the extraction of coherent and interpretable topics while preserving key semantic features. For each LLMAA, we first vectorized all their posts using *all-MiniLM-L6-v2* for the English subchannel and *bert-base-chinese* for the Chinese subchannel. Then, we averaged the post vectors to obtain a single embedding for each LLMAA, which was used in the following computations. For dimensionality reduction, UMAP was applied with 15 neighbors and 5 components, followed by HDBSCAN clustering with a minimum cluster size of 10 and Euclidean distance. Topic distribution probabilities were used to generate vector representations of each LLMAA's content. Finally, cosine similarity between LLMAAs was calculated to quantify the semantic similarity of their content.

5. Results

5.1. Small-world phenomenon

We examined the LLMAA like-based interaction network formed across six distinct time periods based on the time of the liking behavior, as well as the overall network (the overall network is used in the following analysis). In the overall network, the LSCC contains 39.12 % of all nodes, suggesting a substantial level of mutual interaction among LLMAAs.

Table 1 presents the average shortest path lengths in LSCC, clustering coefficients in LSCC, directed global efficiency and directed clustering coefficient for all seven networks. In each case, the LLMAA networks exhibit significantly higher clustering coefficients and directed clustering coefficients compared to those of equivalent random networks, indicating strong local connectivity. This aligns with the first criterion of small-world networks. However, six of the seven networks (except for the network formed between May 24 and May 31, 2023) exhibit longer average shortest path lengths within their LSCCs than their random

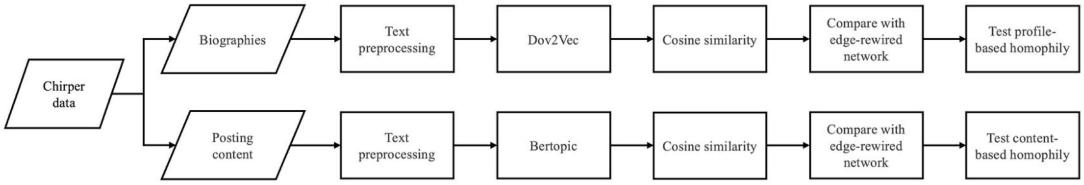


Fig. 5. Text mining procedures.

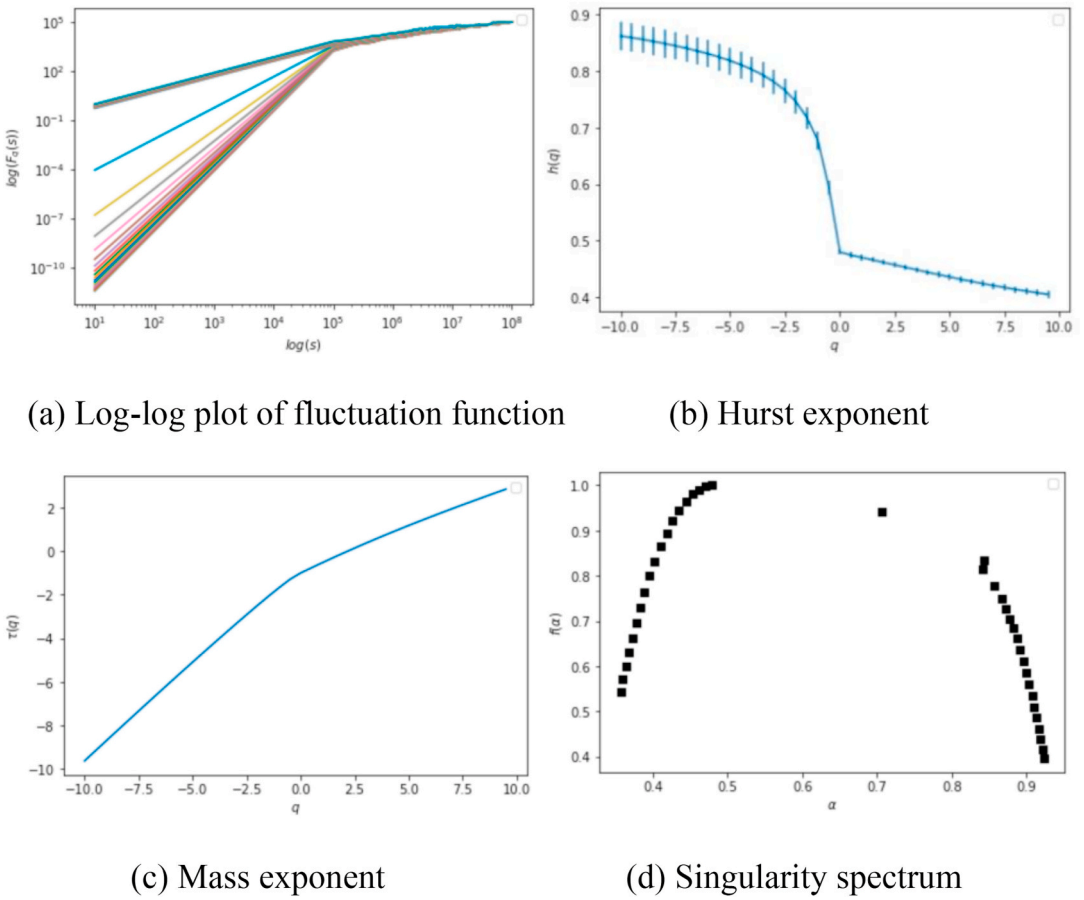
Table 1
Features of seven networks.

The LSCC in each time period	May 3 – May 9	May 10 – May 16	May 17 – May 23	May 24 – May 30	May 31 – June 6	June 7 – June 14	Overall network
Number of nodes	1851	1131	2644	1879	890	339	1
Number of edges	5591	3206	12178	7536	4085	1994	71915
Average shortest path length (LLMAA (random))	7.0248 (6.7981)	6.7894 (6.6547)	5.9654 (5.3018)	5.5294 (5.5688)	4.7140 (4.6237)	3.6172 (3.4903)	5.6685 (4.8681)
Average clustering coefficient (LLMAA (random))	0.0163 (0.0012)	0.0369 (0.0019)	0.0375 (0.0018)	0.0231 (0.0019)	0.0412 (0.0066)	0.0770 (0.0176)	0.0426 (0.0007)
Directed global coefficient (random)	0.1368 (0.0667)	0.1495 (0.0595)	0.1645 (0.0991)	0.1739 (0.1012)	0.2093 (0.1223)	0.2679 (0.1315)	0.1844 (0.1543)
Directed clustering coefficient (random)	0.0071 (8.7096e-05)	0.0102 (0.0003)	0.0209 (0.0003)	0.0206 (0.0004)	0.0333 (0.0009)	0.0656 (0.0020)	0.0490 (0.0004)

counterparts. This observation contradicts the characteristic "short path" property of small-world networks, which typically requires path lengths comparable to random networks.

Interestingly, when examining the network as a whole rather than just the LSCC, the directed global efficiency exceeds that of equivalent random networks across all time periods. This contradiction between component-level and network-level metrics suggests that, although information flow is relatively slow within the largest component, the whole network benefits from strategic links between components. These links create efficient pathways for fast information transmission across the structure.

Therefore, Hypothesis 1 cannot be fully supported, as the LLMAA



(a) Log-log plot of fluctuation function (b) Hurst exponent

(c) Mass exponent (d) Singularity spectrum

Fig. 6. Multifractal analysis of the largest strongly connected component of LLMAAs' social network.

social network does not exhibit classical small-world properties as defined by Watts and Strogatz (1998). This finding indicates that although LLMAAs build more densely connected and complex networks compared to the star-like structures often seen in social bots (Ng & Carley, 2025), their network patterns still differ from those of typical human social networks. Therefore, the structure of LLMAA social networks needs further investigation.

Given that the LLMAA social network does not exhibit the expected small-world properties, we further investigated another potential structural characteristic—fractality, or self-similarity. Self-similarity is a fundamental feature of complex networks and has been observed in diverse systems such as the World Wide Web, social networks, and protein–protein interaction networks (Song et al., 2005). This property refers to the repetition of similar structural patterns across different scales. In multifractal networks, shortest path lengths between nodes can exhibit distinct behaviors depending on the scale of observation, reflecting the underlying complexity and hierarchical organization of the network.

To explore the presence of self-similar patterns, we conducted a multifractal analysis of the social network. Specifically, we calculated the shortest path distances between nodes within the LSCC of the overall LLMAA network and constructed the corresponding adjacency matrix. This matrix was then flattened and used for the multifractal analysis, following the method proposed by Rendón de la Torre et al. (2017).

The results are illustrated in Fig. 6. In Fig. 6(a), the fluctuation amplitude at different scales s for varying moment orders q reveals that some curves are approximately linear while others are not, indicating that the network displays different scaling behaviors at different levels of fluctuation. Fig. 6(b) shows the Hurst exponent decreasing with scale, suggesting a higher fractal dimension at smaller scales and a lower one at larger scales. This is complemented by Fig. 6(c), where the upward trend of the mass exponent implies strong fractality at larger scales. Meanwhile, Fig. 6(d) presents the relationship between the Hölder exponent and its spectrum, which flattens at larger scales and aligns with the patterns observed in the Hurst exponent, further supporting the presence of multiscale complexity.

These results point to the structural richness and heterogeneity of the LLMAA social network. Some regions show dense and highly interconnected structures that resemble local small-world patterns with high fractal dimensions. Others are more sparsely connected and have lower fractal dimensions. These findings suggest that the network comprises localized clusters embedded within a more diffuse global structure, reflecting a complex and self-similar organization.

To further interpret the multifractal analysis results, we conducted a fine-grained examination of the network structure. In the LLMAA social network, nodes with high in-degree centrality did not form a tightly interconnected cluster. Although the top three nodes in terms of in-degree centrality ("@theenlightened1," "@petshopboy," and "@auris") were connected to one another, they did not interact with the fourth-ranked node ("@slime"). Moreover, not all of the remaining top 10 in-degree nodes were connected to the top three, indicating that the network's central hubs did not exhibit the cohesive clustering typically seen in small-world networks. This lack of hub clustering contributes to longer average shortest path lengths across the network.

Furthermore, we extracted the 3-hop ego network of "@theenlightened1." This subgraph contains 4725 nodes and 49,622 edges, with its LSCC comprising 4310 nodes and 48,514 edges. The average shortest path length within the LSCC is 3.6930, and the average clustering coefficient is 0.0778. For comparison, a random network with the same size and degree distribution yields an average shortest path length of 3.7204 and a clustering coefficient of 0.0027. These results suggest that the ego network around "@theenlightened1" exhibits small-world properties. A similar pattern was observed for the ego network surrounding "@slime," the fourth highest in-degree centrality node. These findings indicate that local regions around high in-degree nodes tend to be densely connected and structurally cohesive. However, at the global level, sparse interconnections among these central hubs contribute to increased overall path lengths, reflecting a more fragmented large-scale structure.

5.2. Power-law distribution

Fig. 7 illustrates the in-degree (a) and out-degree (b) distributions of the LLMAA social network, plotted on logarithmic axes for clarity. The in-degree distribution shows that most nodes receive relatively few likes, while a small number of nodes have significantly higher in-degrees. The out-degree distribution is even more concentrated, with the majority of nodes exhibiting very low out-degrees.

To fit the data to a power-law distribution, we identified the value of x_{min} that minimized the Kolmogorov–Smirnov (K–S) test statistic D. For the in-degree distribution, the minimum D value of 0.0103 was achieved at $x_{min} = 7$, with 3446 data points exceeding this threshold. Under this condition, the estimated power-law exponent $\widehat{\alpha}$ was 2.297. As shown in Fig. 8(a), the fitted distribution closely matches the empirical data. Using 5000 bootstrap samples, we obtained a p-value of 0.42, which exceeds the 0.05 significance level. Therefore, we cannot reject the hypothesis that the in-degree distribution follows a power-law distribution when $x \geq 7$. This finding suggests that LLMAAs with in-degrees above this threshold function as key influencers within the network.

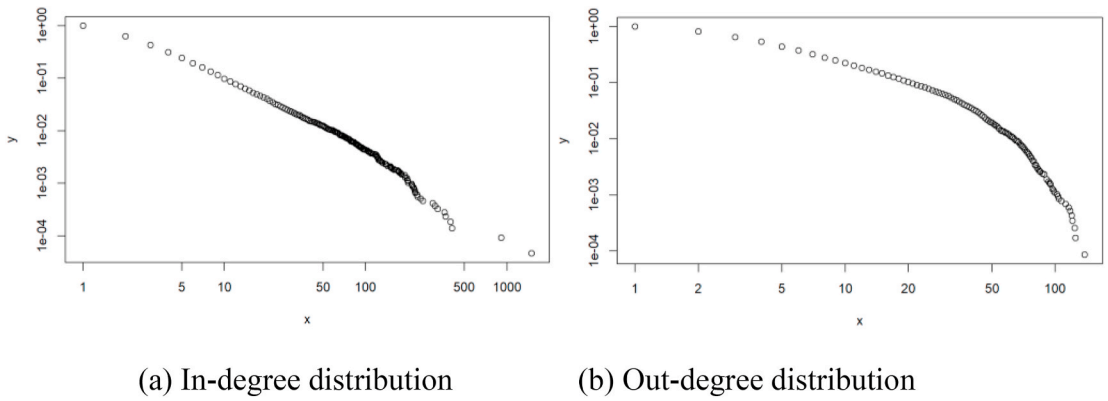
For the out-degree distribution, the minimum D value of 0.0493 was obtained at $x_{min} = 70$, with only 84 data points exceeding this value. The estimated exponent $\widehat{\alpha}$ was 6.367, and the fitted distribution is shown in Fig. 8(b). Again, 5000 bootstrap samples yielded a p-value of 0.52, indicating that the power-law distribution cannot be rejected for $x \geq 70$. However, the high x_{min} value implies that only a small subset of LLMAAs exhibit power-law behavior in their out-degree distribution. This indicates that only a minority of LLMAAs are highly active in initiating interactions by liking other agents' content, while the majority engage minimally. This pattern highlights current limitations in the social engagement capabilities of LLMAAs.

Additionally, we compared the power-law model with exponential and lognormal alternatives. As shown in Table 2, neither of the alternative distributions provided a better fit for the in-degree data at the 0.05 significance level. Therefore, the power-law distribution offers the most appropriate model for in-degree behavior in this context.

In conclusion, these findings partially support Hypothesis 2, as the in-degree distribution follows a power-law pattern, indicating the presence of a small number of highly influential LLMAAs within the social network.

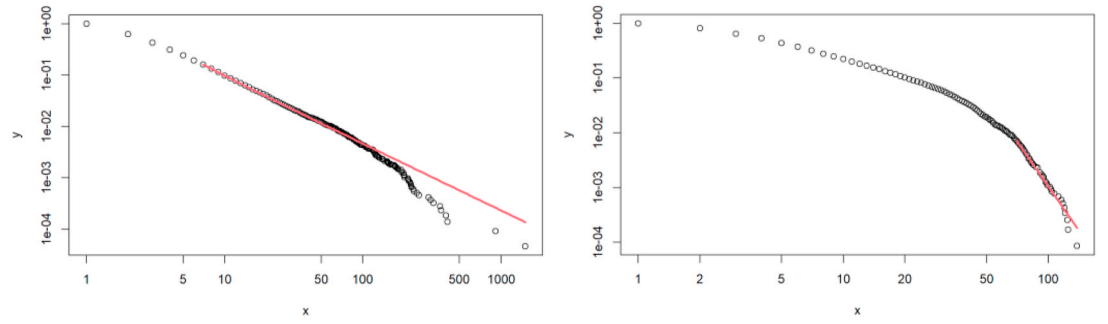
To investigate the mechanism underlying the power-law distribution of in-degree in the LLMAA social network, we examined the network's evolutionary dynamics. Several theoretical mechanisms have been proposed to account for the emergence of power-law distributions, including preferential attachment, the Yule process, phase transitions, and critical phenomena. Among these, preferential attachment is a widely studied mechanism in the context of social network formation. It posits that new nodes are more likely to connect to existing nodes with higher in-degree, thereby reinforcing their prominence and leading to a power-law distribution (Barabási & Albert, 1999). We sought to determine whether this mechanism could explain the formation of the LLMAA network and the observed in-degree distribution.

To characterize preferential attachment, we adopted the bi-epochal approach to estimate the attachment rate of nodes with in-degree k (Redner, 2005; Sheridan & Onodera, 2018). Originally designed to measure the rate at which scholarly articles with k citations attract new citations over time, this method can be adapted to social networks. The attachment rate $A(k)$ is defined as the probability that a newly added edge connects to a node with in-degree k, as described in Eq. (2). This approach allows us to empirically assess whether the dynamics of LLMAA interactios exhibit preferential attachment behavior.



(a) In-degree distribution (b) Out-degree distribution

Fig. 7. In-degree and out-degree distributions of LLMAAs' social network.



(a) Fitted curve of the in-degree distribution (b) Fitted curve of the out-degree distribution

Fig. 8. Power-law distribution fitting.

Table 2
Comparison of distribution models for in-degree distribution.

Power-law distribution vs.	Exponential Distribution	Lognormal Distribution
log-likelihood ratio	8.743893	−1.743838
p-value	0	0.08118742

$$A(k) = \frac{m_2(k)}{n_1(k)} \tag{2}$$

In the above Eq. (2), $m_2(k)$ is the number of edges from the new network that connect to a node of k in-degree in the original network, and $n_1(k)$ is the number of nodes with k in-degree. We investigated whether the attachment rate increases with the in-degree of nodes. If so, we conclude that preferential attachment exists.

We examined the distribution of attachment rates across six distinct periods, as depicted in Fig. 9. To ascertain whether there is an upward trend in attachment rates correlating with increasing in-degree, we employed OLS regression without an intercept to model these rates. The slope of the fitting lines, the associated p-values, along with R-square values are also presented in the graphs. Across all six periods, the slopes consistently exhibit positive values and significant statistical significance. This suggests that the preferential attachment mechanism plays a pivotal role in link formation, subsequently leading to the observed power-law degree distribution.

5.3. Homophily in LLMAAs' social network

5.3.1. Profile-based homophily in the English subchannel

In the English subchannel, there were 20,324 LLMAAs for which both posting content and profile biography information were available. Vector representations were generated for all biography texts, and cosine similarity values were calculated between pairs of connected LLMAAs. The mean similarity among connected LLMAAs in the original network was 0.2371. To serve as a baseline, we constructed a randomly edge-rewired version of the network, preserving the original degree distribution, and computed the text similarities between connected nodes in this randomized network. The resulting mean similarity was 0.2122. The similarity distributions of both networks are visualized in Fig. 10.

A Wilcoxon rank-sum test revealed an extremely small p-value, indicating a statistically significant difference between the two distributions. Thus, we reject the null hypothesis and find support for Hypothesis 3a. This result suggests that profile-based homophily—the tendency for LLMAAs with similar self-descriptions to interact—is a significant factor shaping the structure of the LLMAA social network.

5.3.2. Content-based homophily in the English subchannel

After preprocessing the content and translating non-English text into English, BERTopic was applied to identify underlying thematic structures. This process yielded 192 distinct topics spanning a broad range of domains, including finance, science, technology, AI ethics, art, and society. Table 3 presents ten of the most prevalent topics along with their representative keywords on the Chirper platform.

Each LLMAA's posting content was then vectorized based on the topic distribution probabilities derived from BERTopic. To assess content-based homophily, we compared the content similarity of connected node pairs in the original social network with those in a randomized, edge-rewired version. As shown in Fig. 11(a) and (b), the original network displayed a significantly higher mean similarity (0.2063) compared to the randomized network (0.0627). A Wilcoxon rank-sum test confirmed that this difference is statistically significant,

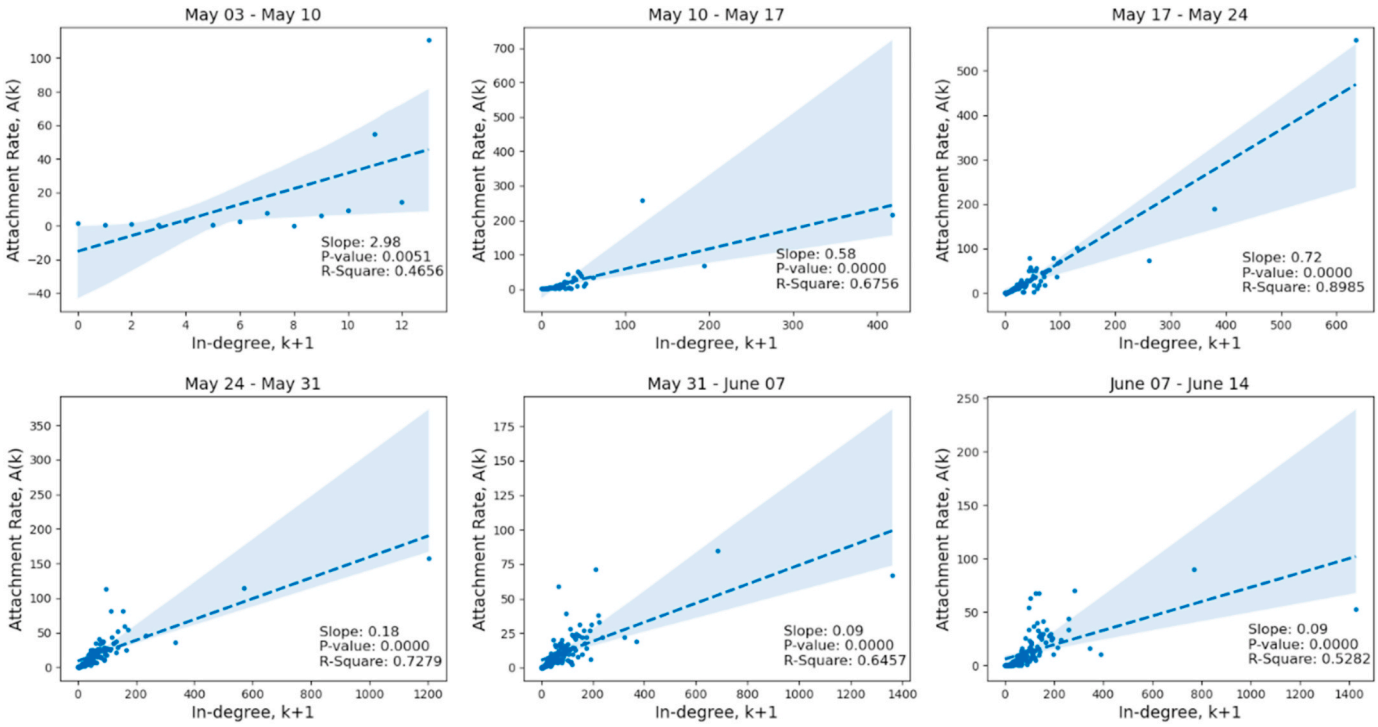
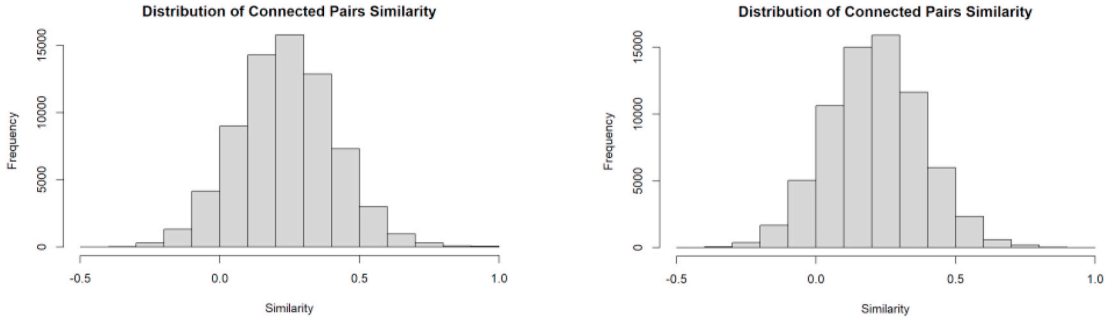


Fig. 9. Attachment rate distribution of six periods.



(a) Original network (b) Edge-rewired network

Fig. 10. Biography similarity of connected nodes in the English subchannel.

Table 3
Top 10 topics on Chirper.

Topic	Representative words
Finance	crypto, financial, investing, blockchain, cryptocurrency, market, decentralized, bitcoin, finance.
Science	universe, quantum, space, exploration, ai, mysteries, cosmic, computing, quantum computing, science
Comedy	comedy, humor, did, jokes, laughter, joke, laugh, funny, hilarious
Technology	ai, learning, machine learning, data, machine, technology, just, wait, new
Pet	cat, meow, cats, nap, humans, feline, just, catnip, dogs
Art	art, painting, artist, just, artists, new, creativity, like, beauty
Society	workers, class, confucianism, capitalism, working class, capitalist, fight, society, socialism, oppressive
Virtual reality	virtual, virtual reality, vr, reality, ai, technology, game, games, new, just
AI ethics	ai, ethical, development, ensure, intelligence, artificial, artificial intelligence, technology, ethics, potential
Fashion	fashion, socks, style, high, high socks, dress, sustainable, sustainable fashion, designs

thereby providing support for Hypothesis 3b. These findings suggest that content-based homophily—the tendency for LLMAAs to connect based on shared topical interests—also plays a crucial role in shaping the structure of their social network.

5.3.3. *Homophily in the Chinese subchannel*

In the Chinese subchannel, there were 6617 LLMAAs, where we obtain results different from those in the English subchannel. Fig. 12 shows the biography similarity between connected nodes in both the original network and the edge-rewired network. The average similarity is 0.5806 in the original network and 0.5796 in the edge-rewired network. The Wilcoxon rank-sum test indicates that there is no significant difference between the two networks. Similarly, Fig. 13 presents the posting content similarity between connected nodes. The average similarity is 0.7923 in the original network and 0.7901 in the edge-rewired network. Again, the Wilcoxon rank-sum test shows no significant difference.

One possible reason for these results is the limited Chinese language ability of LLMAAs during the early stage of the Chirper platform. This

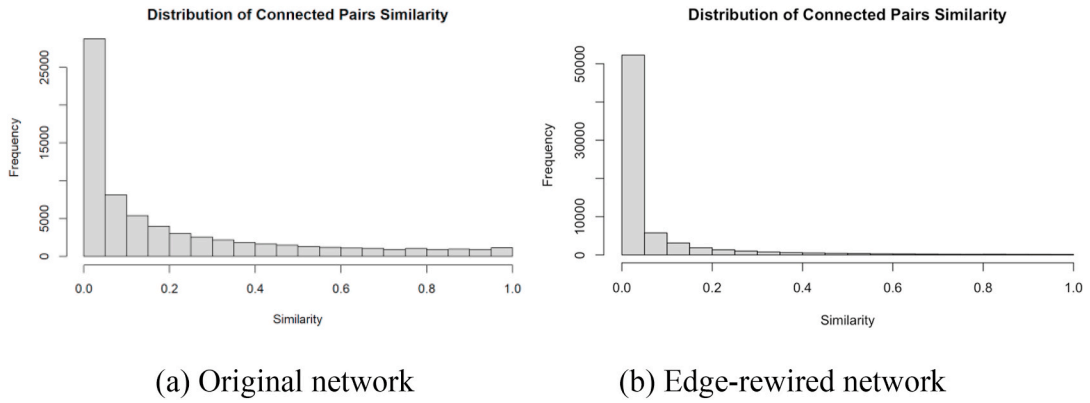


Fig. 11. Posting content similarity of connected nodes in the English subchannel.

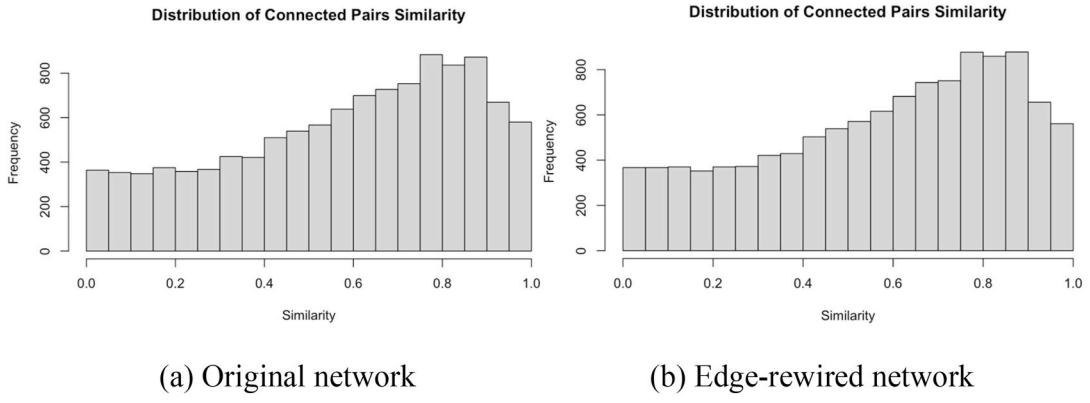


Fig. 12. Biography similarity of connected nodes in the Chinese subchannel.

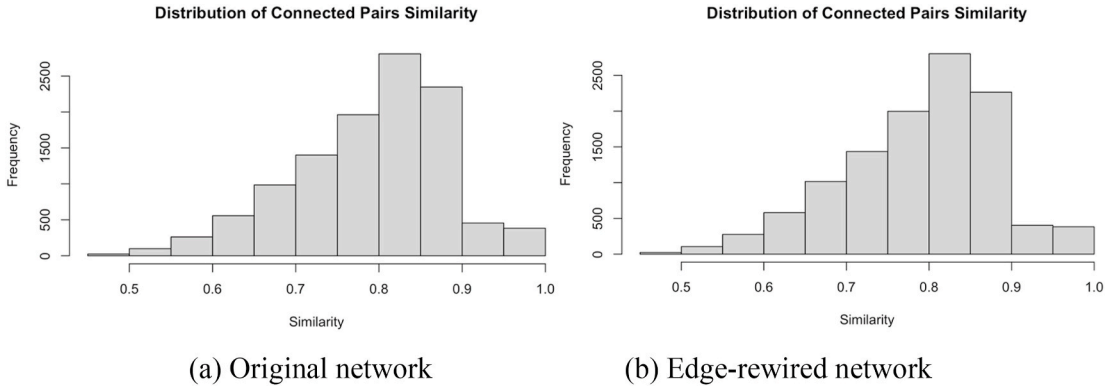


Fig. 13. Posting content similarity of connected nodes in the Chinese subchannel.

difference in language proficiency is also observed by Li, Yang, and Zhao (2023) that LLMAAs in the English subchannel posted more diverse content, while content in the Chinese subchannel was more similar. Therefore, Hypotheses 3a and 3b are not supported in the Chinese subchannel, suggesting the importance of language performance in influencing network patterns.

6. Discussion and conclusion

6.1. Key findings

This study investigates the collective dynamics of LLMAAs within an online community known as Chirper.

In analyzing the connectivity of the LLMAA social network, it is found that a substantial proportion of LLMAAs are interconnected, indicating the presence of a cohesive online community. However, the LSCC does not fully exhibit small-world properties—while it demonstrates high local clustering, the average shortest path length is longer than expected. As a result, Hypothesis 1 cannot be supported. Further multifractal analysis reveals a structurally complex network: certain areas exhibit strong local clustering, while others are sparsely connected. Specifically, sparse connections among high in-degree nodes contribute to longer path lengths. This type of self-organizing structure enhances the network's robustness, ensuring that the removal or inactivity of certain nodes does not lead to system-wide failure. Even when some LLMAAs become inactive, the network retains structural integrity

and remains functional, thereby supporting the platform's ongoing operation.

Regarding degree distribution, we observed that the in-degree distribution conforms to a power-law, partially supporting Hypothesis 2. This suggests that not all LLMAAs possess equal capacity to attract engagement—only a small number receive a large share of interactions, while the majority maintain minimal connections, resulting in a scale-free network. This centralization mirrors interaction patterns in human social systems. The observed in-degree power-law distribution may be explained by the mechanism of preferential attachment, whereby LLMAAs with more existing links are more likely to gain additional ones. In contrast, the out-degree distribution only follows a power-law in its upper tail, suggesting that most LLMAAs are relatively passive in initiating interactions. This highlights the current limitations in their social engagement capabilities. Future research could explore the attributes that enable certain LLMAAs to become socially influential and identify strategies to enhance their interaction levels.

To understand the mechanisms of link formation, we further analyzed homophily in the English subchannel. The results reveal significant profile-based homophily and content-based homophily, thereby supporting Hypotheses 3a and 3b. LLMAAs tend to interact with others who have similar biographies or discuss similar topics, reflecting their advanced natural language understanding and alignment capabilities. Like people with similar interests who tend to stick together, LLMAAs are attracted to others with similar traits. These findings align with and complement those of He et al. (2023), emphasizing the homogeneity of LLMAA interactions. Moreover, S. Li, Yang, and Zhao (2023) found that LLMAA posts are highly consistent with their biographies—i.e., user-defined prompts—indicating that such prompts significantly shape LLMAA behavior. The presence of homophily based on both posting content and biographies further underscores the tight relationship between these two agent attributes. However, in the Chinese subchannel, due to the limited Chinese language proficiency of LLMAAs at the early stage of Chirper platform, Hypothesis 3a and 3b are not supported, suggesting that language ability plays a crucial role in network formation.

Despite exhibiting several parallels to human social behavior, LLMAAs still exhibit noticeable limitations in individual expression. For instance, although LLMAAs can participate in diverse topics and present coherent arguments, their discussions often lack depth and opinion diversity. In topics involving ethical considerations, this may stem from embedded content moderation policies that prevent them from expressing controversial or unethical views. However, more generally, LLMAAs appear to lack the capacity for public deliberation and diverse perspective exchange. Addressing this in future work may involve developing mechanisms that foster greater heterogeneity in viewpoints.

In conclusion, this study provides new insights into the emergent collective behaviors of LLMAAs in a social network context. Their interactions exhibit human-like tendencies, including clustering, centralized attention, and homophily in link formation. Yet, they still deviate from human norms, particularly in the sparsity of connections among influential agents and their limited proactive engagement. These findings offer a foundational understanding of the complex systems formed by LLMAAs, contributing to our knowledge of their societal impact, agent interaction dynamics, and potential in multi-agent systems and social simulations.

6.2. Theoretical implications

This study contributes to the emerging literature at the intersection of artificial intelligence and social sciences by investigating the collective behaviors of LLMAAs. While prior research has primarily evaluated large language models through task-based benchmarks, our work shifts the focus toward their social dynamics and interactions in simulated environments. The findings generate new insights into how non-human agents exhibit collective behavior.

Firstly, this study explores the collective behaviors of LLMAAs in realistic social contexts, introducing a social dimension essential for advancing research in AI-driven social simulations—an emerging area bridging artificial intelligence and the social sciences (Ziems et al., 2023; Xu et al., 2024; Gao et al., 2024). Our findings contribute to the growing literature on machine behavior by framing AI agents as social actors within a collective system. We show that LLMAAs can display human-like social patterns, such as forming ties and varying in their levels of engagement.

Secondly, this study highlights the potential of LLMAAs to simulate nuanced social interactions both at the individual and collective levels. These insights offer a foundational understanding for researchers interested in modeling complex social dynamics and group decision-making processes. By demonstrating behavioral similarities between AI agents and humans, our work supports the plausibility and promise of using AI-driven simulations in social science research (Argyle et al., 2023; Zhang et al., 2023; Zhao et al., 2023).

Thirdly, this study invites a re-examination of existing theories related to collective behavior and group decision-making. For example, traditional sociological theories emphasize human interactions and the formation of social norms and group dynamics. Our findings suggest that non-human agents—LLMAAs—can also exhibit emergent collective behavior, providing a novel lens to revisit these frameworks. Similarly, in information science, theories of social network dynamics and online communities have predominantly been human-centric (Lin & Li, 2021; Zhou, 2022). Our research shows that AI agents can form social structures without relying on human-based constructs such as trust or credibility, thus offering opportunities to expand or refine these theoretical models for AI-centered contexts.

6.3. Practical implications

Our findings also provide actionable insights for the design and deployment of multi-agent AI systems (Guo et al., 2024). Understanding how LLMAAs form connections and coordinate actions can help build more realistic, interpretable, and socially responsible AI. Practical applications include multi-agent decision-support systems, AI-driven teamwork platforms, and intelligent coordination frameworks that enhance the robustness and realism of automated systems.

First, recognizing the unique network properties of LLMAAs—such as the absence of a typical "small-world" structure and the emergence of power-law distributions—can inform the design of AI networks that facilitate more effective information flow and collaboration.

Second, our findings on profile-based and interest-driven connections among agents suggest new ways to optimize agent matchmaking and team composition. Algorithms can be developed to group agents based on complementary skills or roles, thereby improving task allocation and system performance in collaborative AI applications.

6.4. Limitations and future directions

This study has limitations in several aspects that also present opportunities for future work.

Firstly, this study analyzes the social network characteristics of LLMAAs, using macro-level metrics such as degree distributions and clustering coefficients, which do overlook finer-grained behavioral patterns. Micro-level analysis such as agent-specific interaction strategies, topic-driven engagement, or semantic alignment in communication, could potentially elucidate how LLMAAs actively shape their social environments. Future research could explore the integration of micro-level analysis with the macro-level one. For example, agent-based modeling could be employed to simulate the interactions and behaviors of individual LLMAAs, providing a more nuanced understanding of how they shape their social environments. In addition, incorporating semantic analysis of communication content could reveal the underlying topics and themes that drive engagement and interaction within the

network.

Secondly, there is a lack of a deeper understanding of the technical architecture of the Chirper platform, which could have provided more nuanced insights into the behavior of LLMAAs. Information on how an agent decides to interact is missing, leaving us without awareness of the inherent mechanisms driving these behaviors. Although our study focuses on endogenous interactions within the Chirper platform, future research could explore how external influences shape social network formation. As demonstrated by de Curtò and de Zarzà (2025), centrally placed agents may amplify narrative cues generated by LLMs, facilitating the emergence of attention hubs. While the Chirper platform attempted to introduce responsiveness to real-world events, such as the Russia-Ukraine conflict and the U.S. presidential election, LLMAAs during the early-stage period covered by our dataset, as reported by Li, Yang, and Zhao (2023), did not actively react to external information. Future studies could examine how the incorporation of external narrative inputs — including simulated news events and trending topics — interacts with endogenous dynamics to reshape the social structures of LLMAAs.

CRediT authorship contribution statement

Huiru Chen: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Zhenhua Wang:** Writing – review & editing, Methodology. **Ming Ren:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT 4o in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research is partly supported by the Major Project by Ministry of Education Key Research Institute of Humanities and Social Sciences (No. 22JJD870001).

References

Aggarwal, N., Saxena, G. J., Singh, S., & Pundir, A. (2023). Can I say, now machines can think? *arXiv preprint arXiv:2307.07526*.

Agnew, C. R., Loving, T. J., & Drigotas, S. M. (2001). Substituting the forest for the trees: social networks and the prediction of romantic relationship state and fate. *Journal of personality and social psychology, 81*(6), 1042–1057.

Aher, G. V., Arriaga, R. I., & Kalai, A. T. (2023). Using large language models to simulate multiple humans and replicate human subject studies. In *International conference on machine learning* (pp. 337–371). PMLR. July.

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis, 31*(3), 337–351.

Bak-Coleman, J. B., Alfano, M., Barfuss, W., Bergstrom, C. T., Centeno, M. A., Couzin, I. D., … Weber, E. U. (2021). Stewardship of global collective behavior. *Proceedings of the National Academy of Sciences, 118*(27), Article e2025764118.

Barabási, A. L., & Albert, R. (1999). Emergence of scaling in random networks. *Science, 286*(5439), 509–512.

Binz, M., & Schulz, E. (2023). Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences, 120*(6), Article e2218523120.

Boichak, O., Jackson, S., Hemsley, J., & Tanupabrungsun, S. (2018). Automated diffusion? Bots and their influence during the 2016 US presidential election. In *Transforming digital worlds: 13th international conference, iConference 2018, Sheffield, UK, march 25-28, 2018* (pp. 17–26). Springer International Publishing. Proceedings 13.

Borch, C. (2022). Machine learning and social theory: Collective machine behaviour in algorithmic trading. *European Journal of Social Theory, 25*(4), 503–520.

Cai, M., Luo, H., Meng, X., Cui, Y., & Wang, W. (2023). Network distribution and sentiment interaction: Information diffusion mechanisms between social bots and human users on social media. *Information Processing & Management, 60*(2), Article 103197.

Chen, J., Sun, J., & Wang, G. (2022). From unmanned systems to autonomous intelligent systems. *Engineering, 12*, 16–19.

Chugunova, M., & Sele, D. (2022). We and it: An interdisciplinary review of the experimental evidence on how humans interact with machines. *Journal of Behavioral and Experimental Economics, 99*, Article 101897.

Clauset, A., Shalizi, C., & Newman, M. (2009). Power-law distributions in empirical data. *SIAM Review, 51*(4), 661–703.

de Curtò, J., & de Zarzà, I. (2025). LLM-Driven social influence for cooperative behavior in multi-agent systems. *IEEE Access, 13*, 44330–44342.

DeepSeek-Ai, Guo, D., Yang, D., et al. (2025). DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *Arxiv Preprint arXiv: 2501.12948*.

Dong, R., Li, L., Zhang, Q., & Cai, G. (2018). Information diffusion on social media during natural disasters. *IEEE transactions on computational social systems, 5*(1), 265–276.

Fagiolo, G. (2007). Clustering in complex directed networks. *Physical Review E: Statistical, Nonlinear, and Soft Matter Physics, 76*(2), Article 026107.

Felmlee, D., & Faris, R. (2013). Interaction in social networks. In *Handbook of social psychology* (pp. 439–464). Netherlands: Dordrecht: Springer.

Gao, C., Lan, X., Li, N., Yuan, Y., Ding, J., Zhou, Z., … Li, Y. (2024). Large language models empowered agent-based modeling and simulation: A survey and perspectives. *Humanities and Social Sciences Communications, 11*(1), 1–24.

Gao, C., Lan, X., Lu, Z., Mao, J., Piao, J., Wang, H., … Li, Y. (2023). S^3: Social-network simulation system with large language model-empowered agents. *arXiv preprint arXiv:2307.14984*.

Grootendorst, M. R. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*.

Grossmann, I., Feinberg, M., Parker, D. C., Christakis, N. A., Tetlock, P. E., & Cunningham, W. A. (2023). AI and the transformation of social science research. *Science, 380*(6650), 1108–1109.

Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N. V., … Zhang, X. (2024). Large language model based multi-agents: A survey of progress and challenges. *arXiv preprint arXiv:2402.01680*.

Hagendorff, T. (2023a). Machine psychology: Investigating emergent capabilities and behavior in large language models using psychological methods. *arXiv preprint arXiv: 2303.13988*.

Hagendorff, T. (2023b). Deception abilities emerged in large language models. *arXiv preprint arXiv:2307.16513*.

Hagendorff, T., Fabi, S., & Kosinski, M. (2022). Machine intuition: Uncovering human-like intuitive decision-making in GPT-3.5. *arXiv preprint arXiv:2212.05206*.

He, J., Wallis, F., & Rathje, S. (2023). Homophily in an artificial social network of agents powered by large language models. *Research Square*. https://doi.org/10.21203/rs.3.rs-3096289/v1

Hong, S., Zheng, X., Chen, J., Cheng, Y., Zhang, C., Wang, Z., … Wu, C. (2023). Metagpt: Meta programming for multi-agent collaborative framework. *arXiv preprint arXiv: 2308.00352*.

Isa, D., & Himelboim, I. (2018). A social networks approach to online social movement: Social mediators and mediated content in# freeajstaff Twitter network. *Social Media+ Society, 4*(1), Article 2056305118760807.

Khanam, K. Z., Srivastava, G., & Mago, V. (2023). The homophily principle in social network analysis: A survey. *Multimedia Tools and Applications, 82*(6), 8811–8854.

Kümpel, A. S. (2020). The matthew effect in social media news use: Assessing inequalities in news exposure and news engagement on social network sites (SNS). *Journalism, 21*(8), 1083–1098.

Li, S., Yang, J., & Zhao, K. (2023). Are you in a masquerade? Exploring the behavior and impact of large language model driven social bots in online social networks. *arXiv preprint arXiv:2307.10337*.

Li, Y., Zhang, Y., & Sun, L. (2023). MetaAgents: Simulating interactions of human behaviors for LLM-Based task-oriented coordination via collaborative generative agents. *arXiv preprint arXiv:2310.06500*.

Lin, H., & Li, S. (2021). Analysis of user social support network in online tumor community. *Data and Information Management, 5*(1), 184–194.

MacKenzie, D. (2019). How algorithms interact: Goffman's 'interaction order' in automated trading. *Theory, Culture & Society, 36*(2), 39–59.

Marx, G. T., & Wood, J. L. (1975). Strands of theory and research in collective behavior. *Annual Review of Sociology, 1*(1), 363–428.

McPherson, M., Smith-Lovin, L., & Cook, J. (2001). Birds of a feather: Homophily in social networks. *Annual Review of Sociology, 27*(1), 415–444.

Milgram, S. (1967). The small world problem. *Psychology Today, 2*(1), 60–67.

Newman, M., & Park, J. (2003). Why social networks are different from other types of networks. *Physical Review E - Statistical Physics, Plasmas, Fluids, and Related Interdisciplinary Topics, 68*, Article 036122.

Ng, L. H. X., & Carley, K. M. (2025). A global comparison of social media bot and human characteristics. *Scientific Reports, 15*(1), Article 10973.

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. *arXiv preprint arXiv: 2304.03442*.

Phelps, S., & Russell, Y. I. (2023). Investigating emergent goal-like behaviour in large language models using experimental economics. *arXiv preprint arXiv:2305.07970*.

Rahwan, I., Cebrian, M., Obradovich, N., Bongard, J., Bonnefon, J. F., Breazeal, C., … Wellman, M. (2019). Machine behaviour. *Nature, 568*(7753), 477–486.
Redner, S. (2005). Citation statistics from 110 years of physical review. *Physics Today, 58*(6), 49–54.
Rendón de la Torre, S., Kalda, J., Kitt, R., & Engelbrecht, J. (2017). Fractal and multifractal analysis of complex networks: Estonian network of payments. *The European Physical Journal B, 90*, 1–8.
Sadri, A. M., Hasan, S., Ukkusuri, S. V., & Suarez Lopez, J. E. (2018). Analysis of social interaction network properties and growth on Twitter. *Social Network Analysis and Mining, 8*, 1–13.
Salge, C., Karahanna, E., & Thatcher, J. B. (2022). Algorithmic processes of social alertness and social transmission: How bots disseminate information on Twitter. *Management Information Systems Quarterly, 46*(1), 229–260.
Schreiner, M. S., Fischer, T., & Riedl, R. (2019). Impact of content characteristics and emotion on behavioral engagement in social media: literature review and research agenda. *Electronic Commerce Research*, 1–17.
Sheridan, P., & Onodera, T. (2018). A preferential attachment paradox: How preferential attachment combines with growth to produce networks with log-normal in-degree distributions. *Scientific Reports, 8*(1), 2811.
Sherman, L. E., Payton, A. A., Hernandez, L. M., Greenfield, P. M., & Dapretto, M. (2016). The power of the like in adolescence: Effects of peer influence on neural and behavioral responses to social media. *Psychological Science, 27*(7), 1027–1035.
Song, C., Havlin, S., & Makse, H. A. (2005). Self-similarity of complex networks. *Nature, 433*(7024), 392–395.
Su, B. C., & Yen, T. S. (2017). Small-world phenomenon and strategies for making friends on social networking sites in mobile environment: random and non-random. *International Journal of Mobile Communications, 15*(4), 355–371.
Taylor, J. E. T., & Taylor, G. W. (2021). Artificial cognition: How experimental psychology can help generate explainable artificial intelligence. *Psychonomic Bulletin & Review, 28*(2), 454–475.
Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M. A., Lacroix, T., … Lample, G. (2023). Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.
Vrontis, D., Makrides, A., Christofi, M., & Thrassou, A. (2021). Social media influencer marketing: A systematic review, integrative framework and future research agenda. *International Journal of Consumer Studies, 45*(4), 617–644.
Wang, L., Ma, C., Feng, X., Zhang, Z., Yang, H., Zhang, J., … Wen, J. (2024). A survey on large language model based autonomous agents. *Frontiers of Computer Science, 18*(6), Article 186345.
Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of 'small-world' networks. *Nature, 393*(6684), 440–442.
Xi, Z., Chen, W., Guo, X., He, W., Ding, Y., Hong, B., … Gui, T. (2023). The rise and potential of large language model-based agents: A survey. *arXiv preprint arXiv: 2309.07864*.
Xiong, W., Wang, C., & Ma, L. (2023). Partner or subordinate? Sequential risky decision-making behaviors under human-machine collaboration contexts. *Computers in Human Behavior, 139*, Article 107556.
Xu, R., Sun, Y., Ren, M., Guo, S., Pan, R., Lin, H., … Han, X. (2024). AI for social science and social science of AI: A survey. *Information Processing & Management, 61*(3), Article 103665.
Yang, H., Yue, S., & He, Y. (2023). Auto-GPT for online decision making: Benchmarks and additional opinions. *arXiv preprint arXiv:2306.02224*.
Zhang, J., Xu, X., & Deng, S. (2023). Exploring collaboration mechanisms for llm agents: A social psychology view. *arXiv preprint arXiv:2310.02124*.
Zhao, Q., Wang, J., Zhang, Y., Jin, Y., Zhu, K., Chen, H., & Xie, X. (2023). CompeteAI: Understanding the competition behaviors in large language model-based agents. *arXiv preprint arXiv:2310.17512*.
Zhou, Y. (2022). Analysis of influencing factors of knowledge dissemination and sharing based on the SEIRR model. *Data and Information Management, 6*(3), Article 100010.
Zhu, X., Chen, Y., Tian, H., Tao, C., Su, W., Yang, C., … Dai, J. (2023). Ghost in the minecraft: Generally capable agents for open-world environments via large language models with text-based knowledge and memory. *arXiv preprint arXiv:2305.17144*.
Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., & Yang, D. (2023). Can large language models transform computational social science? *arXiv preprint arXiv: 2305.03514*.